

Supervised WSD

The Naïve Bayes Model for Supervised WSD

- The classical approach to WSD that relies on an underlying Naïve Bayes model represents an important theoretical approach in statistical language processing: Bayesian classification (Gale et al. 1992).
- The idea:
 - ✓ The idea of the Bayes classifier (in the context of WSD) is that it looks at the words around an ambiguous word in a large context window. Each content word contributes potentially useful information about which sense of the ambiguous word is likely to be used with it. The classifier does no feature selection. Instead it combines the evidence from all features.
- NaïveBayes is widely used due to its efficiency and its ability to combine evidence from a large number of features. It is applicable if the state of the world that we base our classification on is described as a series of attributes. In our case, we describe the context of the ambiguous word in terms of the words that occur in the context.

- **The Naïve Bayes assumption** is that the attributes used for description are all conditionally independent, an assumption having two main consequences:
 - ✓ all the structure and linear ordering of words within the context are ignored, leading to a so-called “**bag of words model**”;
 - ✓ the presence of one word in the bag is independent of another (which is clearly not true in the case of natural language).

- **Note that:**
 - ✓ In spite of these simplifying assumptions, this model has been proven to be quite effective when put into practice.
 - ✓ The **contextual features** (caratteristici contestuale) used for sense disambiguation here are the content words of the context window themselves.
 - ✓ **The supervised training of the classifier** assumes that we have a corpus where each use of ambiguous words is labeled with its correct sense.

- A *Bayes classifier* applies the *Bayes decision rule* when choosing a class, the rule that minimizes the probability of error. The Bayes decision rule is:

Decide s' if $P(s'/c) > P(s_k/c)$ for $s_k \neq s'$

- Notations:

w - an ambiguous word

$s_1, \dots, s_k, \dots, s_K$ - the senses of the ambiguous word w

$c_1, \dots, c_i, \dots, c_I$ - contexts of w in a corpus

$v_1, \dots, v_j, \dots, v_J$ - words used as features for disambiguation

- With these notations, the Bayes decision rule is:

Decide s' if $P(s'/c) > P(s_k/c)$ for $s_k \neq s'$ (1)

- The Bayes decision rule is optimal because it minimizes the probability of error. This is true because for each individual case it chooses the class (or sense) with the highest conditional probability and hence the smallest error rate. The error rate for a sequence of decisions (for example, disambiguating all instances of w in a multi-page text) will therefore also be as small as possible.

- In formula (1) we usually don't know the value of the probability $P(s_k/c)$, but we can compute it using Bayes' rule:

$$P(s_k / c) = \frac{P(c / s_k)}{P(c)} \cdot P(s_k)$$

- Here:

$P(s_k)$ is the *prior probability* of sense s_k , namely the probability that we have an instance of s_k if we do not know anything about the context. $P(s_k)$ is updated with the factor

$$\frac{P(c / s_k)}{P(c)}$$

which incorporates the evidence (probe) which we have about the context. The result is the *posterior probability* $P(s_k/c)$.

- If all we want to do is choose the correct class, we can simplify the classification task by eliminating $P(c)$ (which is a constant for all senses and hence does not influence what the maximum is). We can also use logs of probabilities to make the computation simpler. Then, we want to assign w to the sense s' where:

$$\begin{aligned}
 s' &= \arg \max_{s_k} P(s_k / c) = \arg \max_{s_k} \frac{P(c / s_k)}{P(c)} \cdot P(s_k) = \arg \max_{s_k} P(c / s_k) \cdot P(s_k) = \\
 &= \arg \max_{s_k} [\log P(c / s_k) + \log P(s_k)]
 \end{aligned}$$

- **Remark:**

This classifier of Gale et al. is an instance of a particular kind of Bayes classifier, the Naïve Bayes classifier. Naïve Bayes is widely used in machine learning due to its efficiency and its ability to combine evidence from a large number of features. It is applicable if the state of the world that we base our classification on is described as a series of attributes. In our case, we describe the context of w in terms of the words v_j that occur in the context.

- **The Naïve Bayes assumption** is that the attributes used for description are all conditionally independent:

$$P(c/s_k) = P\left(\{v_j/v_j \text{ in } c\}/s_k\right) = \prod_{v_j \text{ in } c} P(v_j/s_k) \quad (3)$$

- **Remark:** In our case, the Naive Bayes assumption has two consequences:
 - ✓ all the structure and linear ordering of words within the context is ignored; this is referred to as **a bag of words model**;
 - ✓ the presence of one word in the bag is independent of another (which is clearly not true...)

- With the Naive Bayes assumption, we get the following modified decision rule for classification:

Modified decision rule for classification:

Decide s' if [(3) in (2)]

$$s' = \arg \max_{s_k} \left[\log P(s_k) + \sum_{v_j \text{ in } c} \log P(v_j / s_k) \right] \quad (4)$$

In (4) we compute $P(v_j/s_k)$ and $P(s_k)$ as follows:

Modified decision rule for classification:

Decide s' if [(3) in (2)]

(4)

In (4) we compute $P(v_j/s_k)$ and $P(s_k)$ via Maximum-Likelihood estimation, perhaps with appropriate smoothing (netezire), from the labeled training corpus, and using the following notations:

$C(v_j, s_k)$ = the number of occurrences of v_j in a context of sense s_k in the training corpus;

$C(s_k)$ = the number of occurrences of sense s_k in the training corpus;

$C(w)$ = the total number of occurrences of the ambiguous word w .

- **Computing $P(v_j/s_k)$ and $P(s_k)$ via the Maximum-Likelihood estimation:**

$$P(v_j/s_k) = \frac{C(v_j, s_k)}{\sum_t C(v_t, s_k)} \quad (5)$$

$$P(s_k) = \frac{C(s_k)}{C(w)} \quad (6)$$

- $P(v_j/s_k)$ and $P(s_k)$ are the model parameters.
- (5) and (6) are used in (4).

After estimation, based on the training corpus, of parameters $P(v_j/s_k)$ and $P(s_k)$, the model has been trained and can be used as a classifier for WSD – by determining the most probable sense of an ambiguous word given the context in which it occurs (see the algorithm that follows).

The algorithm for supervised Bayesian disambiguation

1. **comment: training**
2. **for all senses s_k of w do:**
3. **for all words v_j in the vocabulary do**
4.
$$P(v_j/s_k) = \frac{C(v_j, s_k)}{\sum_t C(v_t, s_k)}$$
5. **end**
6. **end**
7. **for all senses s_k of w do:**
8.
$$P(s_k) = \frac{C(s_k)}{C(w)}$$
9. **end**
10. **comment: Disambiguation**
11. **for all senses s_k of w do:**
12. $\text{scor}(s_k) = \log P(s_k)$
13. **for all words v_j in the context window c do:**
14. $\text{scor}(s_k) = \text{scor}(s_k) + \log P(v_j/s_k)$
15. **end**
16. **end**
17. **choose $s' = \arg \max_{s_k} \text{scor}(s_k)$**

- **Conclusion:** the Bayes classifier uses information coming from all words in the context window in order to make the disambiguation decision - at the cost of a somewhat unrealistic independence assumption.
- **Test results:** Gale, Church and Yarowski (1992) report this disambiguation algorithm working correctly on 90% of the occurrences of 6 ambiguous nouns (in the Hansard corpus). The tested nouns are: duty, drug, land, language, position and sentence.